SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Technology Stack

Date	01 JULY 2026
Team ID	xxxxxx
Project Name	OptiCrop – AI Based Crop Recommendation System
Maximum Marks	2 Marks

Technology Stack Details :

S.No	**Architecture Component / Layer**	**Technology Chosen**	**Justification / Purpose**
1	**Frontend / Client-Side**	HTML5, CSS3	Used to create a responsive, user-friendly interface for entering soil and weather parameters and displaying crop recommendations.
2	**Backend / Server-Side**	Python, Flask	Flask handles user requests, processes input data, loads the trained machine learning model, and returns crop predictions efficiently.
3	**Database / Data Storage**	Pickle (.pkl Model File)	The CSV file is used for training the model, while the Pickle file stores the trained machine learning model for fast prediction without retraining

4	**Cloud / Hosting / Deployment**	Render	Render provides simple, free cloud deployment for Flask applications, making the project accessible online. *(If not deployed yet, write **Localhost (Development Environment)**.)*
5	**Version Control & CI/CD**	Git & GitHub	Git tracks code changes, while GitHub enables collaboration, backup, and version management among team members.
6	**Third-Party APIs / Other Tools**	Scikit-learn, Pandas, NumPy, Pickle	Used for data preprocessing, exploratory data analysis, model training, evaluation, visualization, and crop prediction.